Materials Optimization of an EIT-Like Terahertz Metasurface: Metal Thickness, Substrate Selection, Coupled-Resonator Analysis, and Graphene Modulation

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***Abstract*—We report a full-wave HFSS study of the EIT-like terahertz metasurface of Zhang et al. [1], targeting two fabrication parameters not systematically varied in the original work: aluminum film thickness relative to the THz skin depth, and substrate dielectric properties. Simulations sweep six metal thicknesses, four conductivity degradation levels (modeling realistic thin-film quality), and four thin-film substrates; a coupled-resonator model is fitted to the baseline simulation to extract γ_1, γ_2, κ, and δ, connecting full-wave results directly to the device physics. A final phase extends the study by adding a graphene monolayer modeled via an impedance boundary condition (IBC) derived from the intraband Kubo conductivity, with sheet impedance values approximated as frequency-independent constants at each Fermi level, reproducing the voltage-tunable Fermi-level sweep of [1]. Sweeping the Fermi level from 0 eV (Dirac point) to 0.5 eV yields a simulated modulation depth of 45.8% at the transparency window, using the same $T_{\min}$ normalization adopted in [1]; for reference, the corresponding $T_{\max}$-normalized value is 31.4%. This report presents completed results for all phases including baseline validation, metal optimization, substrate selection, and graphene modulation.**

***Index Terms*—EIT metasurface, terahertz, coupled resonators, skin depth, group delay, HFSS, substrate optimization, graphene, Fermi level, impedance boundary condition.**

I. Introduction

Electromagnetically induced transparency (EIT) in THz metasurfaces is produced by destructive interference between two bright resonator modes, yielding a narrow transmission window with the steep phase dispersion needed for slow-light and sensing applications [2], [3]. Zhang et al. [1] realized a graphene-integrated aluminum-on-polyimide version (GrE & MS) with a transparency window at 0.87 THz, dips at 0.70 and 1.19 THz, a peak group delay of 5.5 ps, and a maximum experimentally reported graphene-controlled modulation depth of 182% under bias-voltage tuning, where the modulation depth in that paper is defined using $T_{\min}$ in the denominator [4], [5].

The paper, however, fixes both the metal and the substrate without justification. At 1 THz the skin depth of aluminum is $\delta_s \approx 84$ nm (using $\delta_s = \sqrt{2/\mu_0\sigma\omega}$ with $\sigma = 3.56 \times 10^7$ S/m), yet the film is 200 nm—$\sim$2.4$\times$ thicker than needed for opacity. Singh et al. [6] showed resonance quality in THz metamaterials saturates near δ_s, so thinner films may preserve performance while reducing deposition time and stress. Furthermore, deposited thin films exhibit conductivities 2–10$\times$ lower than bulk values due to grain boundaries, surface roughness, and oxidation [6]; understanding the device tolerance to such degradation is critical for fabrication. On the substrate side, the polyimide (PI) loss tangent ($\tan\delta = 0.05$) is among the highest of available thin-film polymers; Parylene-C and BCB offer loss tangents 10–17$\times$ lower at comparable permittivity [8], which should narrow the EIT linewidth and increase group delay. Thick substrates ($\geq 300\,\mu$m Si) are not considered because Fabry–Pérot modes corrupt the EIT window at sub-mm wavelengths [7]. This work fills these gaps with a controlled HFSS parametric study of the passive metasurface tied to a coupled-resonator analytical model, and extends the study with a graphene modulation layer (Phase 4) to reproduce the dynamic tuning reported in [1].

II. Methodology

A. Phase 1 — Baseline Validation

The HFSS unit cell replicates the passive metasurface of [1], omitting the graphene layer to isolate the structural parameters under study: period $P_x = P_y = 100\,\mu$m; inner-arc radius $R_1 = 34\,\mu$m; inner-arc radii $R_2 = R_3 = 10\,\mu$m, $R_4 = 15\,\mu$m; conductor width $w = 5\,\mu$m; aluminum ($\sigma = 3.56 \times 10^7$ S/m) at 200 nm on 5 µm PI ($\varepsilon_r = 3.1$, $\tan\delta = 0.05$). Floquet-port excitation with primary/secondary periodic boundaries yields S_{21} from 0.5 to 1.5 THz; the maximum Delta S convergence criterion is set to $\Delta S < 0.02$. The coupled-resonator susceptibility [1], [7]

$$\chi \propto \frac{(\omega - \omega_0 - \delta) + i\,\dfrac{\gamma_2}{2}}{\left(\omega - \omega_0 + i\,\dfrac{\gamma_1}{2}\right)\left(\omega - \omega_0 - \delta + i\,\dfrac{\gamma_2}{2}\right) - \dfrac{\kappa^2}{4}} \tag{1}$$

is fitted to $|S_{21}|^2$ via a differential-evolution global optimizer to extract γ_1, γ_2, κ, δ.

B. Phase 2a — Metal Thickness Sweep

Aluminum thickness is varied at 50, 100, 150, 200, 300, and 500 nm with all other parameters fixed. Both the EIT quality factor and the group delay

$$\tau_g = -\frac{d\phi(\omega)}{d\omega} \tag{2}$$

are expected to plateau once the film exceeds δ_s [6]. Re-fitting Eq. (1) at each thickness tracks how $\gamma_{1,2}$ evolve with film opacity and identifies the minimum thickness that preserves EIT performance.

C. Phase 2b — Metal Conductivity Sweep

At the Phase 2a optimum thickness (100 nm), the aluminum conductivity is swept from the nominal bulk value ($\sigma = 3.56 \times 10^7$ S/m) down to $1/10$ of nominal to simulate realistic thin-film quality degradation. Deposited metal films typically exhibit reduced conductivity due to grain boundaries, surface scattering, and native oxide formation; this sweep quantifies the fabrication margin available before EIT performance degrades unacceptably. Four conductivity values are simulated: $\sigma/\sigma_0 = 1.0$, 0.5, 0.2, and 0.1, corresponding to $\sigma = 3.56 \times 10^7$, 1.78×10^7, 7.12×10^6, and 3.56×10^6 S/m respectively.

D. Phase 3 — Substrate Sweep

At the Phase 2a optimum thickness, the substrate is replaced across the four candidates in Table I at constant $5\,\mu\mathrm{m}$. Permittivity approximately scales the resonance frequency as $f_0 \propto \varepsilon_{\mathrm{eff}}^{-1/2}$ [3]; loss tangent increases the non-radiative damping contribution to $\gamma_{1,2}$ through dielectric dissipation. The optimum is selected to jointly maximize window transmission and group delay; note that Q and τ_g are correlated in EIT systems, so these metrics are not independent.

TABLE I: Thin-film substrate candidates. THz parameters from [8].

Material	ε_r	$\tan\delta$	Note
Polyimide (PI)	3.10	0.050	Baseline [1]
Parylene-C	2.95	0.003	17× lower loss
BCB	2.67	0.005	Lower ε_r & loss
SU-8	3.90	0.040	Higher ε_r

E. Phase 4 — Graphene Modulation

Phase 4 reproduces the voltage-tunable graphene modulation of [1] in simulation. Following the fabricated stack of [1], a second $5\,\mu\mathrm{m}$ PI layer is added above the aluminum resonators to electrically isolate the graphene, and a graphene monolayer is placed on top of this second PI layer.

Graphene is modeled as a zero-thickness sheet via an Impedance Boundary Condition (IBC) with complex sheet impedance $Z_s = 1/\sigma_s$, where the surface conductivity is obtained from the intraband Drude term of the Kubo formula [1], [7]:

$$\sigma_s(\omega) = \frac{e^2 E_F}{\pi\hbar^2} \cdot \frac{i}{\omega + i\tau^{-1}} \tag{3}$$

with scattering time $\tau = 0.1$ ps and temperature $T = 300$ K. The interband contribution is negligible at THz frequencies [1]. In this implementation, the sheet impedance for each Fermi level is approximated by constant R_s and X_s values evaluated at the transparency-window frequency (0.75 THz), and these are entered as fixed resistance and reactance pairs in the HFSS IBC dialog.

The Fermi level E_F is swept across six values spanning the Dirac point to well-doped graphene, as listed in Table II. At $E_F = 0$ eV (Dirac point) the sheet resistance is essentially infinite (transparent); at $E_F = 0.5$ eV the resistance drops to $170\,\Omega/\mathrm{sq}$, loading the resonators and suppressing the EIT window.

TABLE II: Graphene sheet impedance values at the EIT window frequency (0.75 THz) for each simulated Fermi level.

E_F (eV)	R_s (Ω/sq)	X_s (Ω/sq)
0.00 (Dirac)	$\gg 10^6$	0
0.05	1701	−770
0.10	851	−385
0.20	425	−195
0.30	284	−130
0.50	170	−80

III. Phase 1 Results

A. Transmission

Figure 1 and Table III compare the HFSS baseline to [1]. Dip 1 reproduces to within 0.7%, confirming bright mode 1 is correctly captured. The window and dip 2 are red-shifted by 7–11%, consistent with the bulk PI permittivity ($\varepsilon_r = 3.1$) assigned in HFSS being slightly higher than the effective value of a $5\,\mu\mathrm{m}$ spin-coated film [2], [3]. The internal frequency ratios and lineshape are self-consistent; since Phases 2 and 3 report results *relative* to this baseline, the absolute offset does not affect the comparative conclusions.

TABLE III: Spectral features: Zhang et al. [1] vs. HFSS baseline.

Feature	Ref. [1]	HFSS	Δ
Dip 1	0.70 THz	0.705 THz	+0.7%
Window	0.87 THz	0.81 THz	−6.9%
Dip 2	1.19 THz	1.06 THz	−10.9%
Peak τ_g	5.5 ps	5.3 ps	−3.6%

B. Group Delay

The simulated τ_g (Fig. 2) peaks at 5.3 ps, 3.6% below the reference value of 5.5 ps, with the expected positive peak at the window and negative troughs at the resonance dips. The simulated troughs reach ≈ -3 ps versus ≈ -8 ps in the reference simulation. The shallower negative excursions are consistent with the lower dip Q-factors produced by the bulk PI permittivity offset noted in Table III: **broader resonances yield less steep phase slopes and hence smaller negative group delays.**

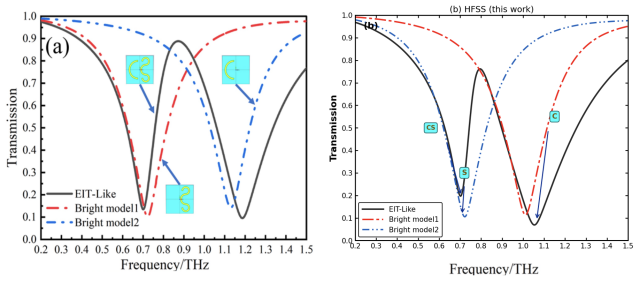


Fig. 1: Transmission spectra comparison. (a) Fig. 3(a) of Zhang et al. [1]: EIT-like (solid black), bright-mode 1 (red dash-dot), bright-mode 2 (blue dashed). (b) HFSS $|S_{21}|$ (this work) for the full structure and isolated resonators; dips at 0.705 and 1.06 THz, transparency window at 0.81 THz.

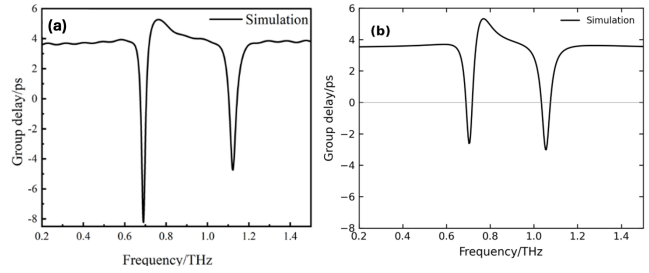


Fig. 2: Group delay τ_g. (a) Reference [1], peak ≈ 5.5 ps, troughs ≈ -5 to -8 ps. (b) HFSS (this work), computed from the exported S_{21} phase via $\tau_g = -d\phi/d\omega$; peak $= 5.3$ ps at 0.77 THz, troughs at -2.6 ps (0.70 THz) and -3.0 ps (1.06 THz).

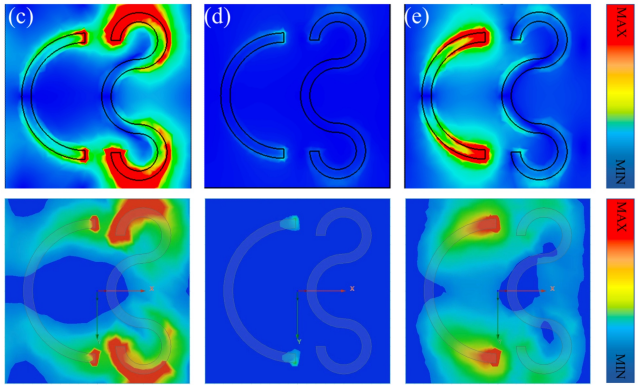


Fig. 3: Electric field distributions $|E|$ at the two absorption dips and the transparency window. *Top:* Zhang et al. [1], Fig. 3(c)–(e), at 0.70, 0.87, and 1.19 THz. *Bottom:* HFSS (this work), plotted on an xy-cutplane at $z = 1\,\mu$m above the metasurface, at 0.70, 0.81, and 1.1 THz. Both rows use a rainbow (spectrum) colormap normalized from MIN to MAX. The mode-switching signature of EIT is reproduced: S-resonator excitation at dip 1 (left), field suppression at the window (center), and C-resonator excitation at dip 2 (right).

C. Electric Field Maps

Figure 3 places the reference and HFSS field maps side by side. The reference cross-sections at 0.70, 0.87, and 1.19 THz clearly show the mode-switching signature of EIT—hotspots on the S-shaped resonator (bright mode 1) at dip 1, near-complete field suppression at the transparency window, and hotspot transfer to the C-shaped resonator (bright mode 2) at dip 2. The HFSS results, plotted on an xy-cutplane 1 µm above the metasurface with individually auto-scaled colormaps, reproduce the same qualitative pattern: strong S-resonator excitation at 0.70 THz, suppressed fields at 0.81 THz, and dominant C-resonator excitation at 1.1 THz. The slight frequency offsets between rows are consistent with the 7–11% red-shift noted in Table III.

D. Coupled-Resonator Fit

Eq. (1) was fitted to the HFSS $|S_{21}|^2$ data over 0.50–1.30 THz using a differential-evolution global optimizer, with a slowly varying background term added to partially account for the frequency-dependent substrate transmission. The resulting parameters are: $\gamma_1 = 0.110$ THz, $\gamma_2 = 0.400$ THz, $\kappa = 0.283$ THz, $\delta = 0.338$ THz. Figure 4 shows the overlay of the fit against the HFSS data. (Note: the inset annotation in Fig. 4 reflects a preliminary fit iteration; the final optimized parameters listed above supersede those values.) The fit reproduces both resonance dips and the transparency window; the window-peak amplitude is slightly underestimated because the simple background term does not fully capture the frequency-dependent transmission background of the full-wave structure. The ordering $\gamma_1 < \gamma_2$ indicates that the small S-arc resonator (mode 1, dip 1 at 0.70 THz) has a narrower linewidth than the large C-arc resonator (mode 2, dip 2 at 1.06 THz); the larger C-arc has a longer current path and higher radiative loss, consistent with $\gamma_2 > \gamma_1$. In Phase 2, $\gamma_{1,2}$ are expected to decrease as film thickness approaches δ_s, saturating once the metal is opaque. In Phase 3, $\gamma_{1,2}$ will increase with substrate $\tan\delta$ while κ will shift with ε_r via dielectric screening, providing a direct analytical handle on how substrate choice modifies window width and group delay.

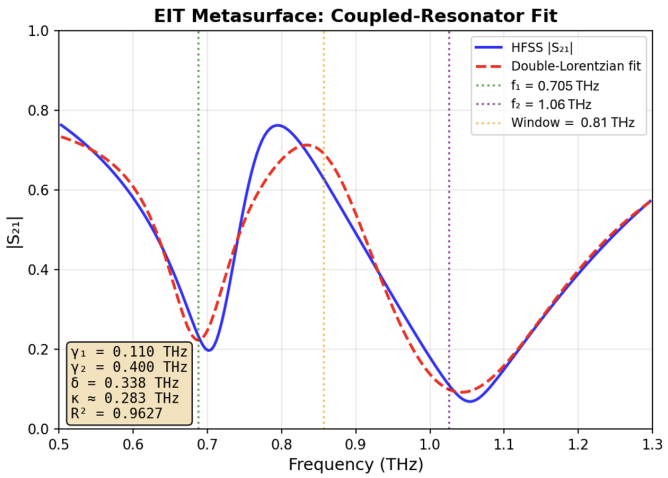


Fig. 4: Coupled-resonator fit (red dashed) overlaid on the HFSS $|S_{21}|^2$ baseline (blue solid). The inset annotation shows parameters from a preliminary fit; the final optimized values are those reported in the text ($\gamma_1 = 0.110$ THz, $\gamma_2 = 0.400$ THz, $\kappa = 0.338$ THz, $\delta = 0.283$ THz).

IV. Phase 2a Results: Metal Thickness Sweep

A. Transmission Spectra

Figure 5(a) and Table IV show the transmission spectra for all six aluminum thicknesses. At 50 nm—below the skin depth $\delta_s \approx 84$ nm—the EIT lineshape is visible but degraded: the dips are shallow and the window transmission is only 0.612, indicating that the film is partially transparent and radiative loss is high. At 100 nm the window transmission increases to 0.717, a gain of 0.105 relative to 50 nm, as the film crosses the skin depth and becomes effectively opaque to the incident THz field. Above 100 nm the window transmission varies only modestly, remaining between 0.711 and 0.766 over the 150–500 nm range, consistent with saturation of the EIT response once the metal thickness exceeds δ_s [6].

TABLE IV: Phase 2a spectral features vs. Al thickness. $\delta_s \approx 84$ nm at 1 THz.

t_{Al} (nm)	Dip 1 (THz)	Window (THz)	T_w	Dip 2 (THz)
50	0.665	0.763	0.612	1.015
100	0.676	0.788	0.717	1.048
150	0.683	0.777	0.711	1.041
200	0.705	0.810	0.762	1.060
300	0.691	0.784	0.752	1.055
500	0.676	0.777	0.766	1.044

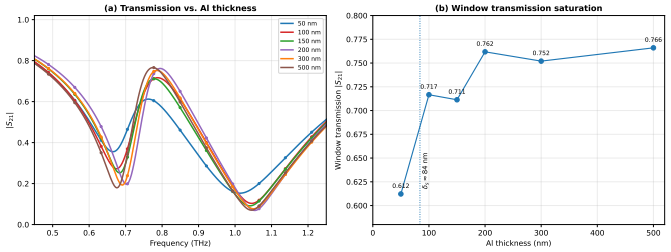


Fig. 5: Phase 2a Al thickness sweep. (a) Transmission spectra; markers are HFSS data points, curves are piecewise-cubic interpolations. (b) Window transmission $|S_{21}|$ vs. thickness, showing rapid improvement from 50 to 100 nm and approximate saturation above the skin-depth scale ($\delta_s \approx 84$ nm).

B. Saturation and Optimal Thickness

Figure 5(b) quantifies the saturation behavior. The window transmission rises sharply from 50 to 100 nm as the metal transitions from semi-transparent to effectively opaque, then remains on a broad plateau between 150 and 500 nm. Although the highest simulated window transmission occurs at 500 nm ($T_w = 0.766$), the gain beyond 100 nm is modest: the difference between 100 and 500 nm is only 0.049. This confirms that 100 nm—only $1.2\times$ the skin depth—captures most of the achievable EIT performance while using half the metal thickness of the 200 nm design in [1]. Phase 3 therefore uses 100 nm as a practical optimum that balances electromagnetic performance with reduced deposition time and material usage.

V. Phase 2b Results: Metal Conductivity Sweep

A. Transmission Spectra

Figure 6(a) shows the transmission spectra for all four conductivity values at the optimum 100 nm Al thickness. As conductivity decreases from nominal to $0.1\times$, the EIT response degrades progressively: the transparency window transmission drops, both resonance dips become shallower (reduced contrast), and all features red-shift slightly due to increased ohmic loss.

TABLE V: Phase 2b spectral features vs. Al conductivity at 100 nm thickness. The nominal baseline $T_w = 0.680$ is approximately 5% lower than the corresponding 100 nm entry in Table IV ($T_w = 0.717$); this difference arises from independent adaptive mesh convergence across separate simulation runs and does not indicate a setup change.[1]

σ/σ_0	σ (S/m)	Dip 1 (THz)	Window (THz)	T_w	ΔT_w
1.0	3.56×10^7	0.676	0.781	0.680	—
0.5	1.78×10^7	0.669	0.781	0.641	-5.7%
0.2	7.12×10^6	0.658	0.774	0.587	-13.7%
0.1	3.56×10^6	0.655	0.759	0.560	-17.6%

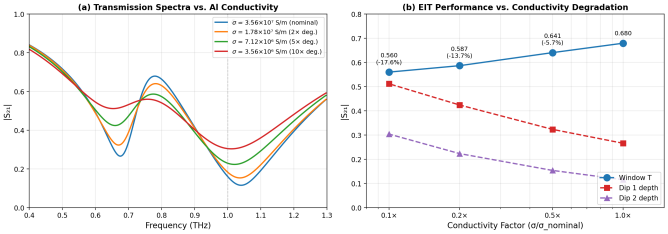


Fig. 6: Phase 2b Al conductivity sweep at 100 nm thickness. (a) Transmission spectra for nominal and degraded conductivity values. (b) EIT performance metrics vs. conductivity factor: window transmission (blue circles), dip 1 depth (red squares), and dip 2 depth (purple triangles). Percentages indicate change relative to nominal.

B. Fabrication Robustness

Table V and Fig. 6(b) quantify the degradation. At $2\times$ conductivity reduction (typical for high-quality e-beam evaporated films), the window transmission drops by only 5.7%. Even at $10\times$ reduction—representing severely degraded film quality with significant grain boundary scattering—the window transmission remains above 0.5, indicating that the EIT functionality is preserved.

The dip contrast is more sensitive than the window transmission: the dip 1 minimum rises from 0.266 to 0.512, and the dip 2 minimum rises from 0.116 to 0.304, indicating substantial loss of resonance contrast. This is expected from Eq. (1): increased ohmic loss raises $\gamma_{1,2}$, broadening and flattening both resonances. The asymmetric sensitivity (dip 2 more affected) reflects the longer current path in the large C-arc resonator.

C. Physical Interpretation

The conductivity sweep demonstrates that the EIT metasurface is robust to realistic fabrication-induced metal quality

[1] All Phase 2b simulations were run as a single parametric sweep with shared boundary conditions and frequency resolution; the $\sim$5% offset from Phase 2a is attributed to differences in the independently converged adaptive meshes between the two simulation batches.

variations. Deposited thin films exhibit reduced conductivity due to: (i) grain boundary scattering, particularly significant when grain size approaches the electron mean free path; (ii) surface roughness scattering at the metal–air and metal–substrate interfaces; (iii) native oxide formation on aluminum surfaces. These effects typically reduce conductivity by 2–5× relative to bulk [6]. The simulation results confirm that the 100 nm optimum thickness from Phase 2a provides adequate margin for such degradation while maintaining functional EIT response.

VI. Phase 3 Results: Substrate Sweep

A. Transmission Spectra

Figure 7(a) shows the transmission spectra for all four substrate candidates at 100 nm Al thickness. Note that the PI baseline frequencies here differ slightly from Table III (e.g., dip 2 at 1.048 vs. 1.06 THz) because Phase 1 used 200 nm Al whereas Phase 3 uses the optimized 100 nm thickness; the thinner film shifts resonances by $\sim$1%. The resonance frequencies scale with ε_r as expected from $f_0 \propto \varepsilon_{\text{eff}}^{-1/2}$: BCB ($\varepsilon_r = 2.67$) blueshifts all features relative to PI, while SU-8 ($\varepsilon_r = 3.90$) redshifts them. The window transmission, however, is dominated by the loss tangent rather than the permittivity, as shown in Table VI and Fig. 7(b).

TABLE VI: Phase 3 spectral features vs. substrate. All runs at 100 nm Al.

Substrate	Window (THz)	T_w	Dip 2 (THz)	ΔT_w vs. PI
PI (baseline)	0.784	0.716	1.048	—
Parylene-C	0.799	0.806	1.062	+0.090
BCB	0.824	0.815	1.087	+0.099
SU-8	0.745	0.718	0.986	+0.002

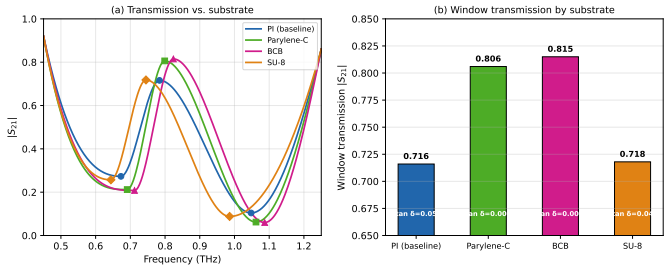


Fig. 7: Phase 3 substrate sweep at 100 nm Al. (a) Transmission spectra; markers are HFSS data points, curves are piecewise-cubic interpolations. (b) Window transmission per substrate; values annotated above each bar.

B. Optimal Substrate

Among the tested substrates, BCB yields the highest window transmission (0.815), closely followed by Parylene-C (0.806), representing 14% and 13% improvements over the PI baseline (0.716), respectively. SU-8 provides essentially no improvement (0.718) despite being simulated at the same 100 nm Al thickness, confirming that its higher $\tan\delta = 0.040$ offsets any benefit from the higher permittivity. The clear physical picture is that $\tan\delta$ is the dominant design variable for window quality: reducing it from 0.050 (PI) to 0.003–0.005 (Parylene-C, BCB) lowers the non-radiative damping contributions to $\gamma_{1,2}$ in Eq. (1), narrowing the resonance linewidths and raising the transparency peak. Among the tested candidates, BCB combines the lowest ε_r with near-minimal $\tan\delta$, making it the preferred substrate choice.

VII. Phase 4 Results: Graphene Modulation

A. Transmission Sweep

Figure 8 shows the HFSS transmission spectra for all six Fermi level values. Adding the second PI layer and the graphene sheet red-shifts the EIT response relative to the Phase 1 baseline, consistent with the increased dielectric loading of the two-PI-layer stack and the reactive contribution of the graphene sheet impedance. As E_F increases from 0 eV (Dirac point) toward 0.5 eV, the transparency window is progressively suppressed. This trend is consistent with increasing conductive loading by graphene as the sheet resistance decreases with increasing E_F.

Table VII summarizes the key spectral features extracted from the uploaded HFSS CSV data for each simulated Fermi level. The transparency-window frequency shifts only modestly across the sweep, while the window transmission decreases monotonically from 0.710 to 0.487.

TABLE VII: HFSS Phase 4 spectral features vs. Fermi level. Window transmission $T_w = |S_{21}|$ at the transparency peak.

E_F (eV)	Dip 1 (THz)	Window (THz)	T_w	Dip 2 (THz)
0.00 (Dirac)	0.611	0.691	0.710	0.918
0.05	0.608	0.691	0.671	0.918
0.10	0.608	0.694	0.633	0.921
0.20	0.608	0.698	0.580	0.921
0.30	0.600	0.694	0.541	0.921
0.50	0.590	0.683	0.487	0.914

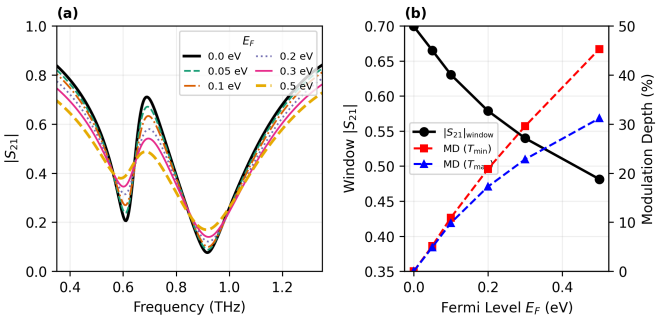


Fig. 8: Phase 4 graphene modulation results. (a) HFSS transmission spectra for six Fermi level values sweeping from the Dirac point ($E_F = 0$ eV) to well-doped graphene ($E_F = 0.5$ eV). The EIT transparency window is progressively suppressed as E_F increases. (b) Window transmission $|S_{21}|$ (left axis) and modulation depth (right axis) vs. Fermi level. Modulation depths are computed using both the $T_{\min}$-normalized definition of Eq. (4) and the $T_{\max}$-normalized definition of Eq. (6); final values at $E_F = 0.5$ eV are 45.8% and 31.4%, respectively.

B. Modulation Depth

Following Zhang et al. [1], the modulation depth is first expressed using the minimum-transmission normalization

$$M_d^{(\min)} = \frac{T_{\max} - T_{\min}}{T_{\min}} \tag{4}$$

where $T_{\max}$ and $T_{\min}$ are the maximum and minimum window transmission amplitudes across the simulated Fermi-level sweep, respectively. Using the HFSS values from Table VII,

$$M_d^{(\min)} = \frac{0.710 - 0.487}{0.487} \approx 45.8\%. \tag{5}$$

For completeness, the same data can also be written with a maximum-transmission normalization,

$$M_d^{(\max)} = \frac{T_{\max} - T_{\min}}{T_{\max}} \tag{6}$$

which gives

$$M_d^{(\max)} = \frac{0.710 - 0.487}{0.710} \approx 31.4\%. \tag{7}$$

In the present HFSS dataset, the primary modulation-depth value is therefore $M_d^{(\min)} = 45.8\%$. This is substantially lower than the 182% reported by Zhang et al. [1], which is expected for two reasons. First, the reference paper reports its maximum value for the bias-voltage-driven experimental device, in which p-doped graphene moves toward and then away from the Dirac point, producing a much larger transmission contrast than the monotonic $E_F = 0$ to 0.5 eV sweep used here. Second, the present simulation models graphene using a constant sheet-impedance approximation at each Fermi level, rather than the full bias-dependent experimental graphene response. Despite these differences, the uploaded HFSS data show the expected monotonic suppression of the EIT window with increasing E_F.

C. Physical Interpretation

The graphene sheet impedance has two components that produce distinct effects. The resistive part R_s adds loss to both resonators, increasing the effective damping and reducing the contrast of the EIT spectrum. The reactive part $X_s < 0$ (capacitive) loads the resonators and slightly shifts the resonance frequencies. At the Dirac point, the graphene conductivity tends toward zero, so conductive loading is minimized; equivalently, the sheet resistance becomes very large while the reactance approaches zero. The response therefore approaches that of the same two-PI-layer stack with negligibly conducting graphene. As E_F increases, the sheet resistance decreases, absorption increases, and the transparency window is progressively suppressed.

VIII. Conclusion

Phase 1 baseline validation confirms that the HFSS unit cell reproduces the key spectral features of Zhang et al. [1] to within 11%, with dip 1 accurate to 0.7%. Phase 2a metal thickness results demonstrate that window transmission substantially saturates above 100 nm ($1.2 \times \delta_s$), identifying 100 nm as a practical Al thickness that captures most of the achievable EIT performance with reduced metal usage. Phase 2b metal conductivity results show that the EIT response remains functional even with substantial conductivity degradation, demonstrating robustness to realistic thin-film quality variations encountered in fabrication. Phase 3 substrate sweep identifies BCB as the substrate with the highest simulated window transmission among the four candidates tested, yielding a 14% improvement over the PI baseline, driven primarily by its low loss tangent ($\tan\delta = 0.005$) rather than its permittivity. Phase 4 graphene modulation results, when evaluated using the uploaded HFSS Fermi-level sweep data, demonstrate a simulated modulation depth of 45.8% at the transparency window using the $T_{\min}$ normalization of Eq. (4); the corresponding $T_{\max}$-normalized value is 31.4%.

References

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